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A spatial model to understand tuberculosis granuloma formation and its impact on disease progression

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Abstract: Tuberculosis (TB) is caused by a bacterium called *Mycobacterium tuberculosis* (Mtb). When Mtb enters inside the pulmonary alveolus, it is phagocytosed by the alveolar macrophages, followed by a cascade of immune responses. This leads to the recruitment and accumulation of additional macrophages and T cells in the pulmonary tissues. A key outcome of this is the formation of granuloma, the hallmark of TB infection. In this paper, we develop a mathematical model of the evolution of granuloma by a system of partial differential equations that is based on the classical Keller–Segel chemotaxis equation. We investigate the effect of different parameters on the formation of granuloma. We present numerical simulation results that illustrate the impact of different parameters. The implication of our result on the disease progression is also discussed.

Keywords: tuberculosis; immune response; chemotaxis model; radial basis function

1 Introduction

Tuberculosis (TB) is one of the world's deadliest infectious diseases. An estimated one-third of the world's population has been exposed to TB. A total of 5–10 % of TB-infected cases develop the active disease while other infected individuals are able to stay in the latent stage. While there has been a large global drop in the number of people newly diagnosed with TB due to the impact of COVID-19, the number of reported cases still stands around 5.8 million. An estimated 1.3 million people were reported killed by the disease in 2020 which was almost double the number caused by HIV/AIDS (0.68 million) [1].

The causative agent of TB is a bacterium called *Mycobacterium tuberculosis* (Mtb), which spreads through the air and develops in the alveoli of the lung. Alveolar macrophages phagocytose the inhaled mycobacteria and transport them into the lung tissues [2]. However, alveolar macrophages are poorly at destroying and clearing the ingested mycobacteria. These macrophages further induce immune response recruiting additional macrophages and T lymphocytes to the site of infection through blood vessels [3], [4]. These cells, as well as other immune cells, will eventually form a well-organized structure called granuloma, the hallmark of pathological TB. The inner cell mass primarily consists of macrophages, infected macrophages, and mycobacteria. It is usually surrounded by a ring of leukocytes and fibroblasts.

The granuloma creates an environment that limits the spread of bacteria. However, it also provides the mycobacterium with a niche in which it can survive for a long period of time, even decades. A dysregulation of the immune system can destroy the granuloma which permits the release of the infectious Mtb into the airways to infect new hosts, or leads to disease progression (secondary TB) in the current host [2].

Early mathematical models for the host-immune dynamics of TB started with the study of co-infection of HIV and Mtb [5]. Some of the earlier models for immune response to TB were systems of ODEs which were used

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to capture the dynamics of the cellular and cytokine network during TB infection [6], [7]. In [7], a large system consisting of three different populations of macrophages, Mtb, Th0, Th1, Th2, and different cytokines was studied numerically. In [6], this model was extended to a two-compartmental model in order to capture the process of cellular activation and priming occurring in the lung and nearby lymph nodes. These models were based on complex mathematical models which were studied by computational simulations with little effort in theoretical analysis.

Some of the recent modeling efforts were able to keep the dimension low [8]–[10]. The model compartments include the principle components: normal macrophages, infected macrophages, T cells and Mtb. These models were rigorously analyzed to study the dynamics of various outcomes. They studied the stabilities of equilibria and the bifurcation phenomena.

However, granulomas are structures that evolve both in time and space, and ODE systems are not sufficient to capture these. In recent years, there have been multiple attempts to model the spatial structure of granulomas. For example, agent-based model (ABM) in [11], [12], hybrid model GranSim in [12], and PDE models [13]. For additional references on computational and mathematical modeling in the immune response to TB, we refer the interested readers to [14]. For a more general review of mathematical modeling of the inflammatory response in lung infections, please see [15].

The main aim of this paper is to develop a spatial-temporal model for granuloma formation, where the model has a lower dimension which allows some meaningful mathematical analysis. For the temporal model, we analyze the stability of the different steady states. We further explored the behavior of the spatial-temporal model system through numerical simulations for both one-dimensional system and two-dimensional system. In the following section, we describe the new model formulation, followed by analysis and numerical simulation of the temporal model in Section 3. In Section 4, we present some numerical simulations for the system in one spatial dimension. In Section 5, we perform numerical solutions of the full chemotaxis model in 2D. Finally, in Section 6, we provide a brief discussion of our article.

2 The model and preliminaries

Mathematical modeling in immunology started as early as the 1960s. Classical models of the immune system began with simple ODE systems, or difference equations, etc. The ODE models consider the evolution of populations of different immune cells and pathogens without considering the spatial diffusion of these cells. We refer the readers to reference [16] for a review of these models. There are few models that consider the motility and chemotaxis of cells in the immune response. One of the earliest PDE models describing immune response was due to Lauffenburger and Keller [17]. In recent years, there are a few spatial-temporal models that investigate the role of immune response in other areas such as tumor cells [18], [19], wound healing [20], [21] and inflammatory response [22]. One of the most recent PDE models of immune response to invading pathogens is due to Su et al. [23]. Their model consisted of over 10 equations that model both the innate and adaptive response. Their model further investigated the fact that increased production of inflammatory cytokines leads to an adaptive response.

Keller and Segel first used partial differential equations to study the interactions of chemotactic cells (slime molds) and a secreted attractant (cAMP) [24], [25]. A typical formulation of the chemotaxis model is the following system:

$$\frac{\partial m}{\partial t} = \nabla \cdot (\mu \nabla m) - \nabla \cdot (\chi m \nabla c) + f(m, c), \quad (1)$$

$$\frac{\partial c}{\partial t} = \nabla \cdot (D \nabla c) + g(m, c). \quad (2)$$

Here, m is the cell density and c is the chemoattractant concentration. $\mu > 0$ and χ are the cell motility and chemotactic coefficient, respectively. $\chi > 0$ corresponds to chemoattraction and $\chi < 0$ corresponds to repulsion. $f(m, c)$ represents cell proliferation and death. The terms in the dynamics of c are random diffusion and

production and/or degradation of chemicals, respectively. Since then, many similar models have been used to study a wide variety of systems that involve chemotaxis. We refer the readers to [26] of a review on this subject.

Our model consists of four partial differential equations that describe the rate of change of several key cells associated with the innate immune response. The bacteria density is represented by $B(t, x)$; the density of normal macrophages is represented by $M(t, x)$; the density of infected macrophages is represented by $M_i(t, x)$; and the density of CD4 T cells is represented by $T(t, x)$.

$$\frac{\partial B}{\partial t} = \underbrace{D_b \Delta B}_{\text{effective diffusion}} + \underbrace{\sigma_b B}_{\text{replication}} - \underbrace{\gamma_b MB}_{\text{engulfment}} + \underbrace{\mu s M_i}_{\text{release}}, \quad (3)$$

$$\frac{\partial M}{\partial t} = \underbrace{D_m \Delta M}_{\text{effective diffusion}} - \underbrace{\nabla \cdot (\chi M \nabla B)}_{\text{effective chemotaxis}} + \underbrace{\beta}_{\text{recruitment}} - \underbrace{\gamma_i MB}_{\text{infection}} - \underbrace{\gamma_m M}_{\text{deactivation}}, \quad (4)$$

$$\frac{\partial M_i}{\partial t} = \underbrace{D_i \Delta M_i}_{\text{effective diffusion}} + \underbrace{\gamma_i MB}_{\text{infected}} - \underbrace{\alpha T M_i}_{\text{necrotized}} - \underbrace{\mu M_i}_{\text{death}}, \quad (5)$$

$$\frac{\partial T}{\partial t} = \underbrace{D_T \Delta T}_{\text{effective diffusion}} - \underbrace{\nabla \cdot (\chi_T T \nabla M_i)}_{\text{effective chemotaxis}} + \underbrace{r T \frac{M_i}{a + M_i}}_{\text{replication of T-cell}} - \underbrace{\delta T}_{\text{death}}, \quad (6)$$

with the Neumann boundary condition

$$\frac{\partial B}{\partial n} = \frac{\partial M}{\partial n} = \frac{\partial M_i}{\partial n} = \frac{\partial T}{\partial n} = 0 \text{ on } \partial\Omega.$$

The first equation is the rate of change of bacteria population where $D_b \Delta B$ represents the diffusion of bacteria. $\sigma_b B$ represents the replication of the bacteria. The term $\gamma_b MB$ represents the phagocytosis of bacteria by macrophages. Intracellular bacteria become extracellular bacteria either by macrophages bursting or by death, denoted by $\mu s M_i$. s is the average total number of bacteria released by a single infected macrophage.

The second equation is the rate of change of macrophages where $D_m \Delta M$ represents the diffusion of macrophages. $\nabla \cdot (\chi M \nabla B)$ represents the chemotaxis toward the bacteria. β is the recruitment rate from the bloodstream. $\sigma_m B$ represents the activation of the macrophages. $\gamma_i MB$ represents the infection of the macrophages, then move to the infected class M_i . $\gamma_m M$ is the natural death of macrophages.

The third equation is the rate of change of the infected macrophages with the extracellular survival of bacteria. The infected macrophages are necrotized by T cells. The term μM_i is the natural death of infected macrophages.

The last equation is the rate of change of CD4 T cells where $D_T \Delta T$ is the effective diffusion of T cells. CD4 T cells are activated by infected macrophages and then move up the gradient of chemoattractants released by infected macrophages. The next term represents the replication rate of T cells in the presence of infected macrophages. Here we choose a log-sigmoidal form of T cell activation to model T cell exhaustion during the presence of infected macrophages. r is the production rate of T cells, and a represents the saturation level of the T cells. When the number of infected macrophages is small, the level of immune response is low. This response increases as the number of infected macrophages increases and is saturated when the number of infected macrophages is sufficiently large. δ is the death rate of T cells.

3 Analysis of the temporal model

In this section, we perform the qualitative analysis of the reduced temporal model by analyzing the existence and stability of boundary equilibria and positive equilibria. Numerical simulations are also carried out to find more interesting dynamical behaviors.

3.1 Analysis of boundary equilibria

It is clear that the system has a unique infection-free equilibrium $E_0 = (0, \frac{\beta}{\gamma_m}, 0, 0)$. Using the survival function method [27], we obtain the following basic reproduction number

$$R_0 = \frac{s\beta\gamma_i + \sigma_b\gamma_m}{\beta\gamma_b}.$$

For the stability of this equilibrium, we have the following theorem.

Theorem 1. *The infection-free equilibrium E_0 is locally asymptotically stable if $R_0 < 1$.*

Proof. The Jacobian matrix at E_0 is

$$\begin{bmatrix} \sigma_b - \frac{\beta\gamma_b}{\gamma_m} & 0 & \mu s & 0 \\ -\frac{\beta\gamma_i}{\gamma_m} & -\gamma_m & 0 & 0 \\ \frac{\beta\gamma_i}{\gamma_m} & 0 & -\mu & 0 \\ 0 & 0 & 0 & -\delta \end{bmatrix}$$

The associated characteristic equation is

$$(\lambda + \delta)(\lambda + \gamma_m) \left[\lambda^2 + \left(\mu + \frac{\beta\gamma_b}{\gamma_m} - \sigma_b \right) \lambda + \mu \left(\frac{\beta\gamma_b}{\gamma_m} - \frac{s\beta\gamma_i}{\gamma_m} - \sigma_b \right) \right] = 0, \quad (7)$$

which is equivalent to

$$(\lambda + \delta)(\lambda + \gamma_m) \left[\lambda^2 + \left(\mu + \frac{\beta\gamma_b}{\gamma_m} - \sigma_b \right) \lambda + \frac{\mu\beta\gamma_b}{\gamma_m}(1 - R_0) \right] = 0. \quad (8)$$

It is then clear that E_0 is asymptotically stable if $R_0 < 1$ and unstable if $R_0 > 1$. $\square$

The immune-free equilibrium is $E_1 = (B^*, M^*, M_i^*, 0)$ satisfies

$$\sigma_b B - \gamma_b M B + \mu s M_i = 0 \quad (9)$$

$$\beta - \gamma_i M B - \gamma_m M = 0 \quad (10)$$

$$\gamma_i M B - \mu M_i = 0 \quad (11)$$

Solving this system yields the following:

$$B^* = \frac{\beta\gamma_b(1 - R_0)}{\sigma_b\gamma_i}, M^* = \frac{\sigma_b}{\gamma_b - s\gamma_i}, M_i^* = \frac{\beta\gamma_b(1 - R_0)}{\mu(\gamma_b - s\gamma_i)}.$$

Hence there is a unique immune-free equilibrium if $R_0 < 1$ and $\gamma_b > s\gamma_i$. If $R_0 > 1$, or $R_0 < 1$ and $\gamma_b < s\gamma_i$, there is no immune-free equilibrium. Let $R_1 = \frac{rM_i^*}{\delta(a + M_i^*)}$.

Theorem 2. *Suppose $R_0 < 1$ and $\gamma_b > s\gamma_i$ if $R_1 < 1$, then E_1 is locally asymptotically stable. Suppose $R_0 < 1$ and $\gamma_b > s\gamma_i$ if $R_1 > 1$, then E_1 is unstable.*

Proof. Suppose that $R_0 < 1$ and $\gamma_b > s\gamma_i$, then the Jacobian matrix at E_1 is given by

$$\begin{bmatrix} \sigma_b - \gamma_b M^* & -\gamma_b B^* & \mu s & 0 \\ -\gamma_i M^* & -\gamma_i B^* - \gamma_m & 0 & 0 \\ \gamma_i M^* & \gamma_i B^* & -\mu & -\alpha M_i^* \\ 0 & 0 & 0 & \frac{r M_i^*}{a + M_i^*} - \delta \end{bmatrix}$$

The characteristic equation is

$$(\lambda - R_1 \delta + \delta)(\lambda + \mu)(\lambda^2 + a_1 \lambda + a_2) = 0,$$

where

$$a_1 = \gamma_i B^* + \gamma_m - \sigma_b + \gamma_b M^* - \mu s \gamma_i M^*$$

$$a_2 = \gamma_b \gamma_m M^* - \sigma_b \gamma_i B^* - \sigma_b \gamma_m - \mu s \gamma_m \gamma_i M^*$$

Therefore, if $R_1 < 1$, we can verify that $a_1 > 0$ and $a_2 > 0$, which means E_1 is locally asymptotically stable. $\square$

3.2 Analysis of positive equilibria

In this section, we explore the existence of interior equilibria and study their stability. A positive equilibrium $E(B, M, M_i, T)$ satisfies:

$$\sigma_b B - \gamma_b M B + \mu s M_i = 0 \quad (12)$$

$$\beta - \gamma_i M B - \gamma_m M = 0 \quad (13)$$

$$\gamma_i M B - \alpha T M_i - \mu M_i = 0 \quad (14)$$

$$r T \frac{M_i}{a + M_i} - \delta T = 0 \quad (15)$$

It follows that

$$M = \frac{\beta}{\gamma_i B + \gamma_m}, \quad M_i = \frac{a \delta}{r - \delta}, \quad T = \frac{\beta \gamma_i (r - \delta) B - \mu a \delta (\gamma_m + \gamma_i B)}{a \alpha \delta (\gamma_m + \gamma_i B)}$$

where B solves the following quadratic equation

$$c_1 B^2 + c_2 B + c_3 = 0,$$

with

$$c_1 = (r - \delta) \sigma_b \gamma_i, \quad c_2 = a \delta \mu s \gamma_i + \beta \delta \gamma_b + r \gamma_m \sigma_b - \beta r \gamma_b - \delta \sigma_b \gamma_m, \quad c_3 = a \delta \mu s \gamma_m.$$

Note that if $r < \delta$, we have $M_i < 0$. Thus we assume $r > \delta$ for the rest of this section. Hence we have the following conclusion

Theorem 3. (a) If $c_2 > 0$, or $c_2 < 0$ and $c_2^2 - 4c_1 c_3 < 0$ there is no positive equilibrium.

(b) If $c_2 < 0$, $c_2^2 - 4c_1 c_3 > 0$, there are two positive equilibria. $B_1 = \frac{-c_2 - \sqrt{c_2^2 - 4c_1 c_3}}{2c_1}$ and $B_2 = \frac{-c_2 + \sqrt{c_2^2 - 4c_1 c_3}}{2c_1}$.

(c) If $c_2 < 0$, $c_2^2 - 4c_1 c_3 = 0$, there is a unique positive equilibrium.

Note that B_1 corresponds to a lower bacteria load, B_2 represents a larger bacteria load in the granuloma.

Remark. A quick calculation reveals that $c_2 < 0$ is equivalent to

$$(r - \delta)(\beta\gamma_b - \gamma_m\sigma_b) > a\delta\mu s\gamma_i.$$

$$\begin{bmatrix} -\frac{\mu s M_i}{B} & -\gamma_b B & \mu s & 0 \\ -\gamma_i M & -\gamma_i B - \gamma_m & 0 & 0 \\ \gamma_i M & \gamma_i B & -\mu - \alpha T & -\alpha M_i \\ 0 & 0 & \frac{raT}{(a + M_i)^2} & 0 \end{bmatrix}$$

3.3 Numerical simulation of the ODE system

In this section, we present numerical simulations of the ODE system. Table 1 lists the parameter values that will be used in this article. These parameters are gathered through various sources such as [3], [7], [14], [23].

For the first set of parameters, we choose the following: $\sigma_b = 0.17$, $\gamma_b = 1.2 \times 10^{-6}$, $\mu = 0.011$, $s = 0.005$, $\beta = 4000$, $\gamma_i = 1.7 \times 10^{-6}$, $\gamma_m = 0.011$, $\alpha = 1.1$, $r = 4.5$, $a = 15,000$, $\delta = 0.33$. For this set of parameters, the system admits the bacteria-free equilibrium $(0, 363636.36, 0, 0)$, the immune-free equilibrium $(10020.76, 142677.30, 220959.07, 0)$, and one positive equilibrium $(10137.78, 141672.03, 1187.05, 1.8599)$. The associated characteristic equation of the Jacobian matrix at the positive equilibrium is

$$p(\lambda) = -0.001832946724 + 0.01164197856\lambda + 0.6807779501\lambda^2 + 2.085130666\lambda^3 + \lambda^4$$

with the following roots

$$\lambda_1 = 0.04181867239, \lambda_2 = -0.07005997897, \lambda_3 = -0.3711168035, \lambda_4 = -1.685772556.$$

For comparison, we increase γ_b to 1.3×10^{-6} . The result is shown in Figure 1.

Our next simulation explores the impact of infection rate γ_i . For this set of simulations, our parameters are chosen to be the following: $\sigma_b = 0.17$, $\gamma_b = 1.008 \times 10^{-6}$, $\mu = 0.011$, $s = 0.005$, $\beta = 4000$, $\gamma_m = 0.011$, $\alpha = 1.1$, $r = 4.5$, $a = 1500$, $\delta = 0.33$. Initial conditions are

$$B(0) = 12, M(0) = 8, M_i(0) = 0, T(0) = 10.$$

We then compare the outcomes for $\gamma_i = 0.9 \times 10^{-4}$ and a slightly larger $\gamma_i = 1.0 \times 10^{-4}$. The results are shown in Figure 2.

Table 1: Parameters used for the numerical simulations.

Parameter	Description	Biological Range	Simulation	Units
D_b	Bacteria diffusion rate	0–0.006	0.004	mm ² /day
D_m	Macrophage diffusion	0.01–0.06	0.04	mm ² /day
D_i	Infected macrophage diffusion	D_m	0.04	mm ² /day
σ_b	Mtb replication rate	0–0.52	0.17	day ⁻¹
γ_b	Bacteria engulfment rate by M	1.25×10^{-9} – 1.25×10^{-7}	1.3×10^{-7}	day ⁻¹
χ_T	Chemotaxis rate of T-cells	0.00565	0.00565	mm ² day ⁻¹
χ	Chemotaxis rate of macrophage	0.00565	0.0045	mm ² day ⁻¹
β	Macrophage recruitment rate	600–1000	1000	day ⁻¹
α	Elimination rate by T cells	0.1–2	1.1	day ⁻¹
γ_m	Macrophage death rate	0.01–0.011	0.0118	day ⁻¹
γ_i	Infection rate	2.5×10^{-11} – 2.5×10^{-7}	1.7×10^{-7}	day ⁻¹
μ	Death of infected macrophage	0.011	0.011	day ⁻¹
r	Maximum T cells replication rate induced by M_i	10^{-8} –10	4.5	day ⁻¹
δ	T Cells death rate	0.33	0.33	day ⁻¹
a	Half-saturation constant for T cell activation	10^4 – 10^5	1500	day ⁻¹

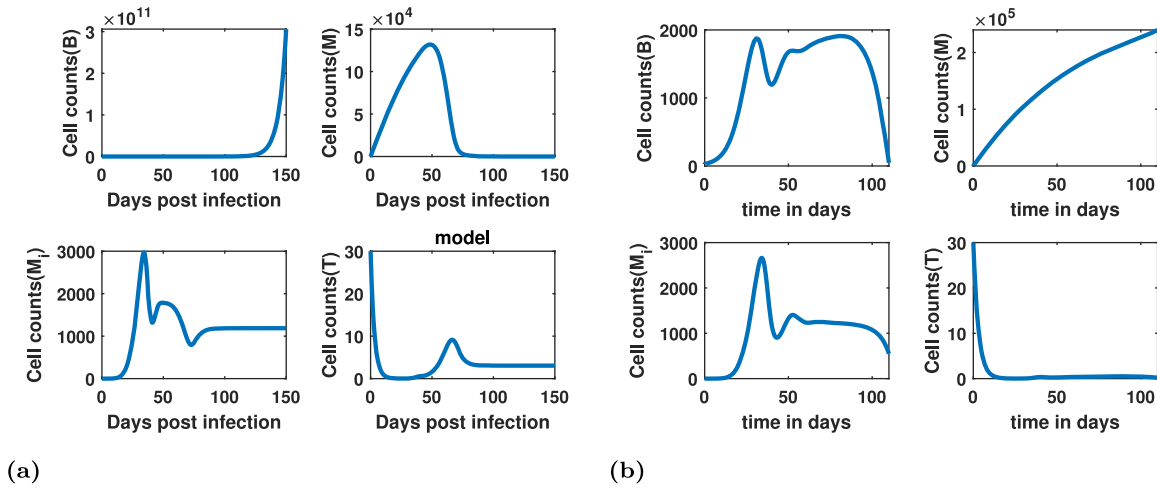


Figure 1: Time evolution of the concentration of cells for different engulfment rates (a) $\gamma_b = 1.2 \times 10^{-6}$; (b) $\gamma_b = 1.3 \times 10^{-6}$.

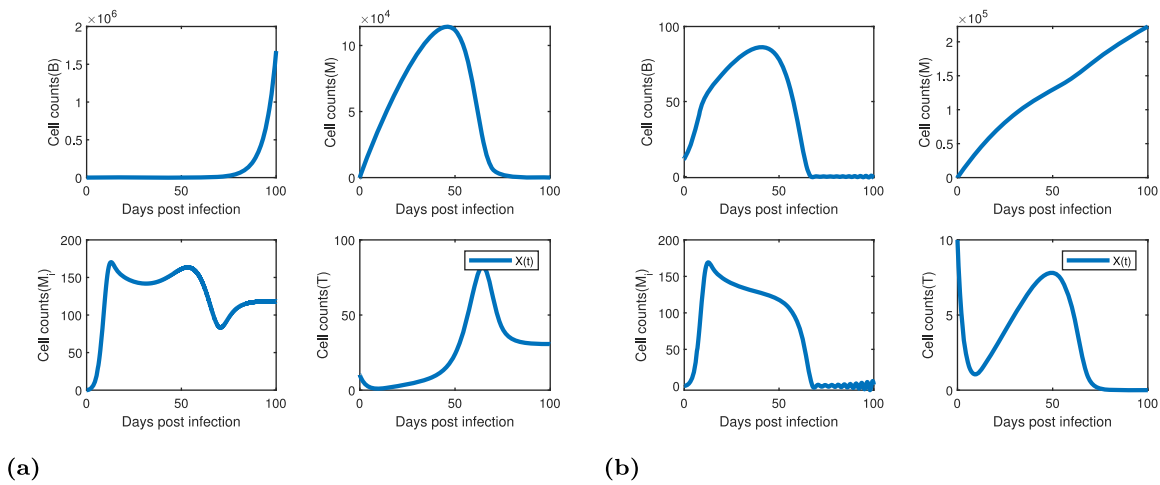


Figure 2: Time evolution of the concentration of cells for different infection rates (a) $\gamma_i = 0.9 \times 10^{-4}$; (b) $\gamma_i = 1.0 \times 10^{-4}$.

Next, we explore the impact of the initial inoculation. For this set of simulations, our parameters are chosen to be the following: $\sigma_b = 0.17$, $\gamma_b = 1.0 \times 10^{-6}$, $\mu = 0.011$, $s = 0.005$, $\beta = 4000$, $\gamma_m = 0.011$, $\gamma_i = 1.0 \times 10^{-4}$, $\alpha = 1.1$, $r = 4.465$, $a = 1500$, $\delta = 0.33$. Initial conditions are

$$B(0) = 13, M(0) = 8, M_i(0) = 0, T(0) = 10.$$

For this condition, the virus will not be able to be controlled by the immune system, see Figure 3. We then decrease the initial inoculation to $B(0) = 12$. For this set of initial conditions, the virus will be able to be cleared out by immune cells.

Finally, we point out that the outcome of the disease is also very sensitive to the expansion rate of T Cells induced by infected macrophages. For this set of simulations, we choose the following set of parameters: $\sigma_b = 0.17$, $\gamma_b = 1.0 \times 10^{-6}$, $\mu = 0.011$, $s = 0.005$, $\beta = 4000$, $\gamma_m = 0.011$, $\gamma_i = 1.0 \times 10^{-4}$, $\alpha = 1.1$, $a = 1500$, $\delta = 0.33$. Initial conditions are

$$B(0) = 12, M(0) = 8, M_i(0) = 0, T(0) = 10.$$

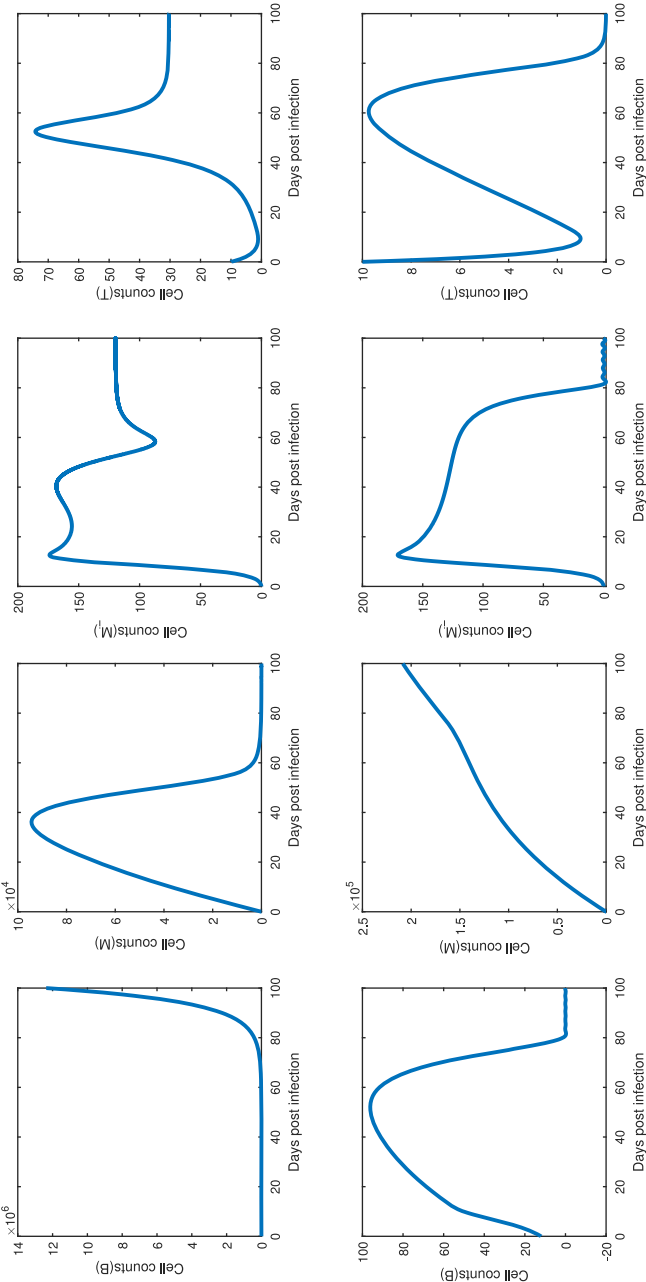


Figure 3: Sensitivity to initial inoculation of the virus. Top panel $B(0) = 13$, bottom panel $B(0) = 12$.

We compare the outcomes for $r = 4.465$ and $r = 4.47$ (Figure 4).

Consider the following set of parameters $\sigma_b = 0.3$; $\gamma_b = 0.12 \times 10^{-5}$, $\mu = 0.011$, $s = 0.005$, $\beta = 8000$, $\gamma_i = 0.00017$, $\gamma_m = 0.011$, $\gamma_i = 0.00017$, $\alpha = 0.5$, $\mu = 0.011$, $r = 3.5$, $a = 15,000$, $\delta = 0.33$. For this set of parameters, the system admits the bacteria-free equilibrium $(0, 72727.72, 0, 0)$ and two positive equilibria

$$(0.1505, 725585.2276, 1561.5142, 0.0018) \text{ and } (123.0926, 250581.4271, 1561.5142, 6.6941)$$

The associated characteristic equation of the Jacobian matrix is

$$p(\lambda) = -0.006272420688 + 0.01155503663\lambda + 1.101315500\lambda^2 + 3.390648844\lambda^3 + \lambda^4$$

with the following roots

$$\lambda_1 = 0.06459246219, \lambda_2 = -0.09733611273, \lambda_3 = -0.3294233391, \lambda_4 = -3.028481854.$$

Thus they are both unstable. For the infection-free equilibrium, $(0, 72727.72, 0, 0)$ is also unstable since $R_0 = 1.0521$. There is no immune-free equilibrium for this set of parameters.

If we alter only the engulfment rate to a smaller value $\gamma_i = 0.000017$. This results in one infection-equilibrium $(0, 72727.72, 0, 0)$, one immune-free equilibrium $(1101.960784, 269058.2960, 458214.4313, 0)$, and a single positive equilibrium $(1234.857832, 250057.9576, 1561.514196, 6.701425866)$. Similar analysis reveals that $R_0 = 0.414583$ and the infection-free equilibrium is asymptotically stable. The positive equilibrium is unstable. The immune-free equilibrium is also unstable.

4 Numerical simulation in one spatial dimension

The model PDE (3)–(6), is solved numerically on a one-dimensional domain of size 5 with zero-flux boundary conditions. In one-dimensional space, the solution exists globally and we need not regularize the chemotaxis term. We used MATLAB's built-in 1D PDE solver PDEPE. The initial conditions are taken as follows

$$B(x, 0) = \begin{cases} 100(1 - e^{-100(x-1)^2}), & \text{for } 0 \leq x \leq 1, \\ 0, & \text{for } 1 < x < 5, \end{cases} \quad (16)$$

$$M(x, 0) = 0, M_i(x, 0) = 0, \quad (17)$$

and

$$T(x, 0) = \begin{cases} 100(1 - e^{-10(x-1)^2}), & \text{for } 1 \leq x \leq 5, \\ 0, & \text{for } 0 \leq x < 1. \end{cases} \quad (18)$$

When the chemotactic strength for T cells is large, the T cells effectively move to the site of infection and the infected macrophages concentration is much lower than the small chemotactic strength. Furthermore, T cells can infiltrate the granuloma with a higher concentration inside (Figure 5). When the chemotactic strength for macrophages is large, the macrophages are aggregated inside the granuloma with a much lower density near the boundary of the granuloma. The concentration of *M. tuberculosis* at the center of the granuloma is also slightly lower due to the availability of macrophages, see Figure 6.

We may also examine the spatiotemporal distribution of macrophages and Mtb when the initial inoculation sites are close. Applying the following initial Mtb load

$$B(x, 0) = \begin{cases} 100(1 - e^{-100(x-1)^2}), & \text{for } 0 \leq x \leq 1, \\ 100(1 - e^{-100*(x-2)^2}), & \text{for } 1.5 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

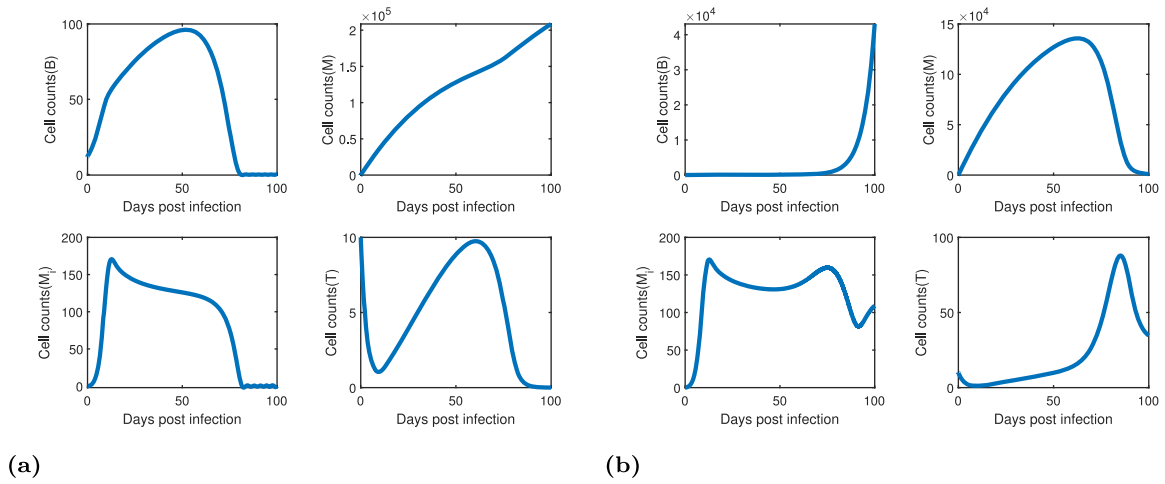


Figure 4: Time evolution of the concentration of cells for different T cell replication rates (a) $r = 4.465$; (b) $r = 4.47$.

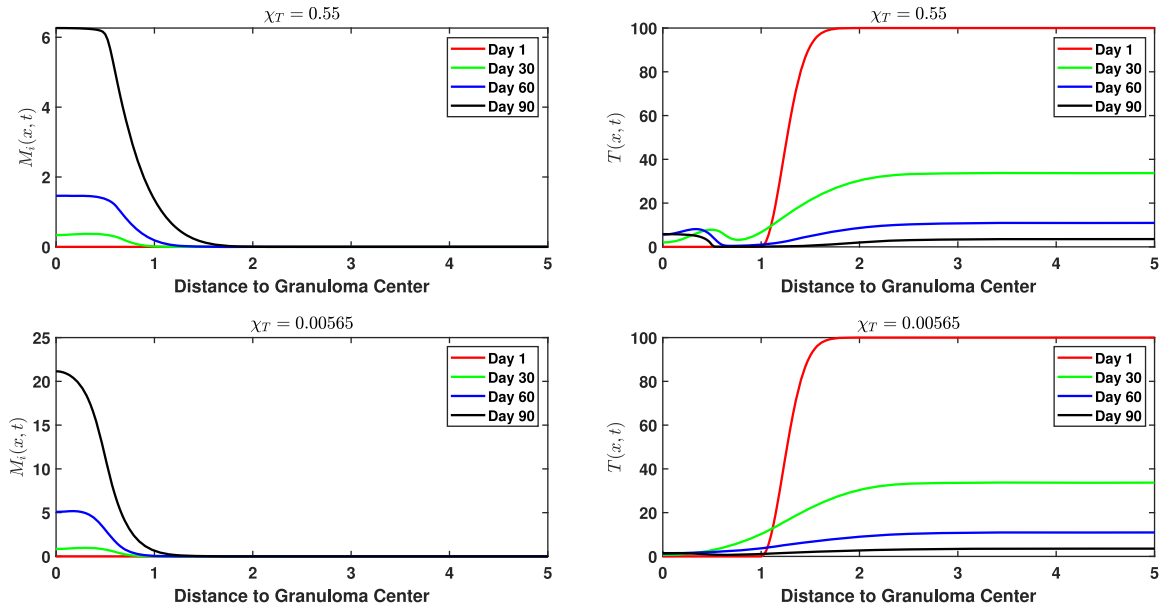


Figure 5: Time evolution of infected macrophages and T cell post-infection with different chemotactic strength for T cells.

we observe that the two individual granulomas may merge and form confluent granulomas. In Figure 7. Histopathological features such as numerous large and confluent granulomas have been shown to be the most useful distinguishing features to differentiate intestinal tuberculosis (ITB) from Crohn's disease (CD) [28], [29].

5 Numerical simulation in two spatial dimensions

Numerical simulation in 2D is carried out based on radial basis function (RBF) method. This is implemented by the formation of a differentiation matrix associated with the spectral method using RBF instead of orthogonal polynomials. The time partial derivatives are discretized by a forward finite difference scheme. For the spatial derivatives, we use RBF $\phi(\|\mathbf{x} - \mathbf{x}_i\|)$ with a set of data $\mathbf{x}_i \in \mathbb{R}^2$, $i = 1, \dots, N$ so that the interpolation of u at time

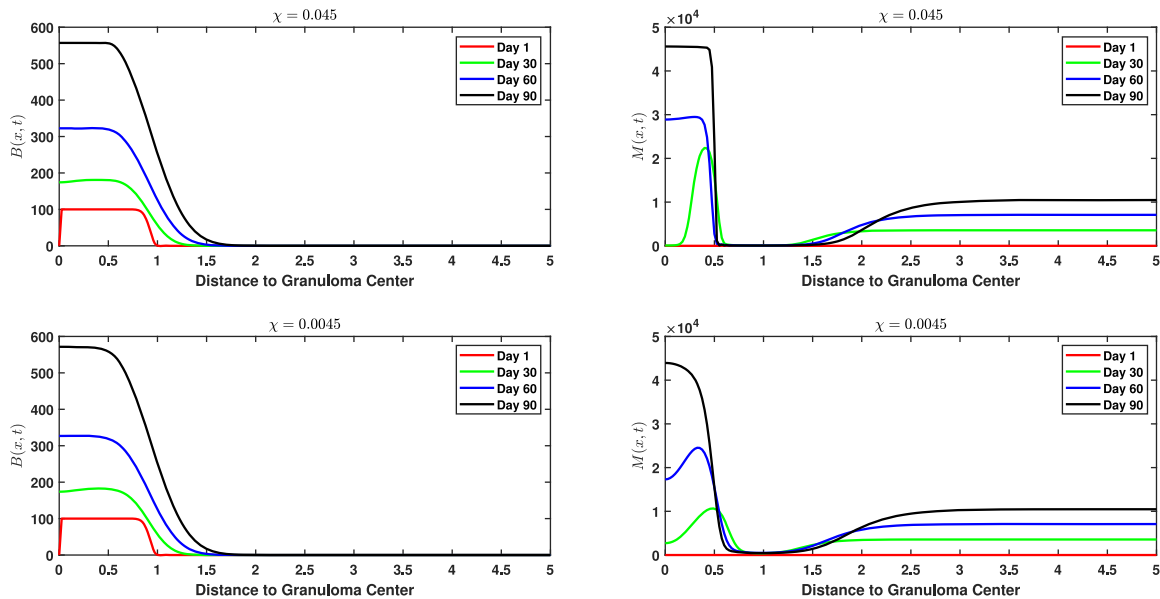


Figure 6: Time evolution of TB, macrophages with different chemotactic strength for Macrophages.

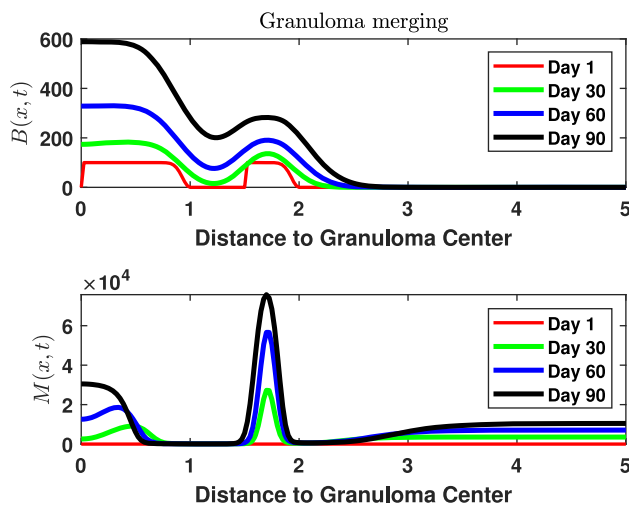


Figure 7: Merging of two granulomas to form a larger confluent granuloma.

$t^n = n\delta t$ can be written as

$$U^n(x) = \sum_{i=1}^N w_i^n \phi(\|\mathbf{x} - \mathbf{x}_i\|).$$

For our numerical simulation, we chose the multiquadric RBF

$$\phi(r) = \sqrt{r^2 + c^2}$$

and the Gaussian RBF

$$\phi(r) = e^{-cr^2}$$

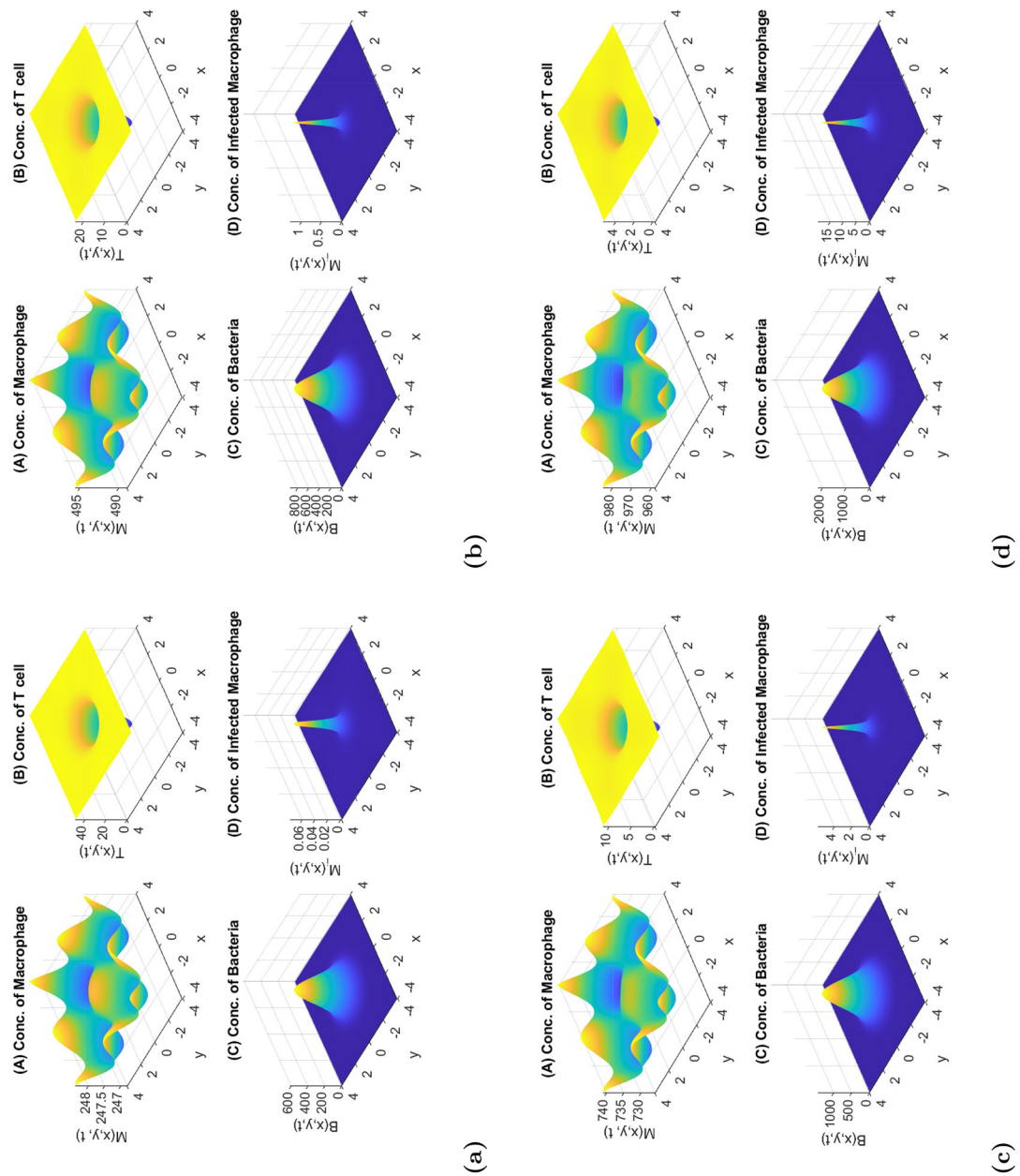


Figure 8: Time evolution of concentration at (a) 2.5 days post-infection, (b) 5 days post-infection, (c) 6.25 days post-infection, (d) 7.5 days post-infection.

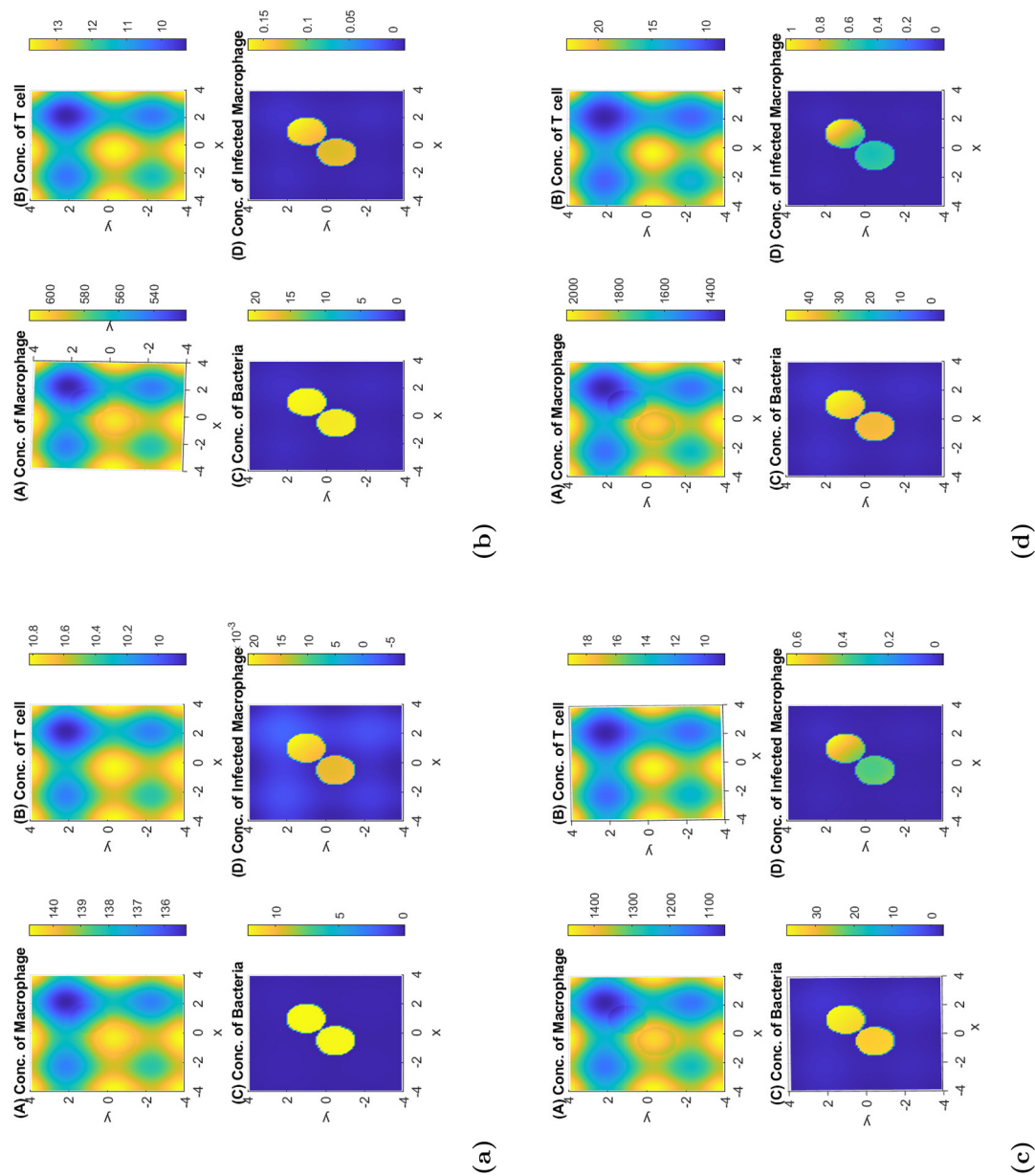


Figure 9: Time evolution of concentration when $\chi = 0.045$ at (a) 2.5 days post-infection, (b) 5 days post-infection, (c) 10 days post-infection, (d) 12.5 days post-infection.

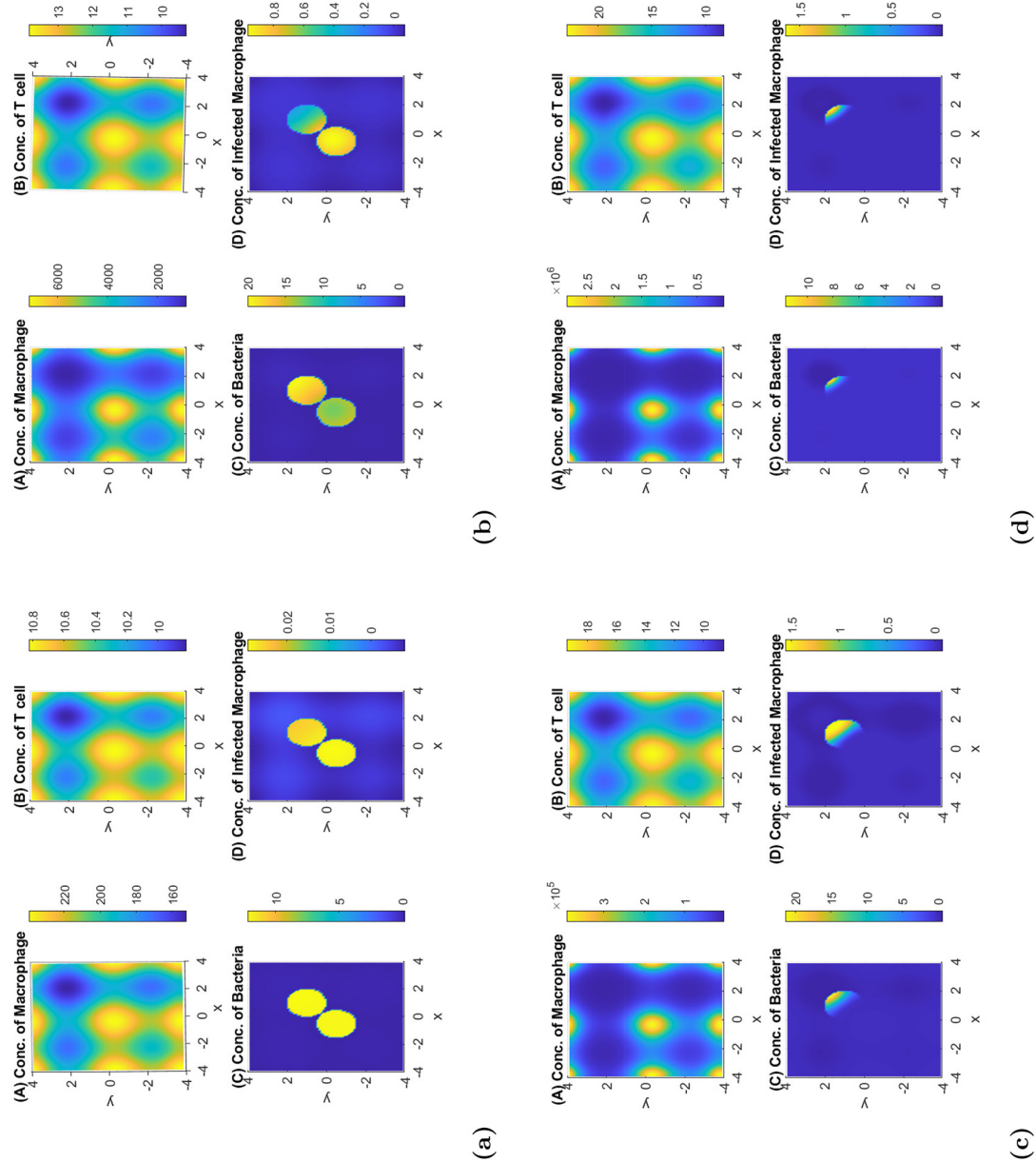


Figure 10: Time evolution of concentration when $\chi = 0.45$ at (a) 2.5 days post-infection, (b) 5 days post-infection, (c) 10 days post-infection, (d) 12.5 days post-infection.

where c is the shape parameter and $r = \|\mathbf{x} - \mathbf{x}_i\|$ is the Euclidian distance between $\mathbf{x}$ and $\mathbf{x}_i$. The derivative of U^n can be computed by differentiating the basis functions, i.e.

$$U_x^n = \sum_{i=1}^N w_i^n \frac{d}{dx} \phi(\|\mathbf{x} - \mathbf{x}_i\|), \quad (20)$$

$$U_{xx}^n = \sum_{i=1}^N w_i^n \frac{d^2}{dx^2} \phi(\|\mathbf{x} - \mathbf{x}_i\|). \quad (21)$$

For simplicity, we use a first-order Euler time integration scheme to deal with the temporal derivatives. So for the following Keller–Segal chemotaxis equations

$$\frac{\partial u}{\partial t} = D_1 \Delta u - \nabla(\chi u \nabla v) + f(u, v), \quad (22)$$

$$\frac{\partial v}{\partial t} = D_2 \Delta v + g(u, v), \quad (23)$$

the approximate solution $\mathbf{U}^n = U^n(x_i, y_j)$, $\mathbf{V}^n = V^n(x_i, y_j)$ at time $t^n = n\Delta t$ and at the collocation points $\{(x_i, y_j)\}_{i,j=1}^N$ can be solved by the following system of vector equations

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \delta t \left(D_1 \left(\mathbf{U}_x x^n + \mathbf{U}_{yy}^n \right) - \chi \left(\mathbf{U}_x^n * \mathbf{V}_x^n + \mathbf{U}_y^n * \mathbf{V}_y^n \right) - \chi \mathbf{U}^n * \left(\mathbf{V}_{xx}^n + \mathbf{V}_{yy}^n \right) \right) + f(\mathbf{U}^n, \mathbf{V}^n) \quad (24)$$

$$\mathbf{V}^{n+1} = \mathbf{V}^n + \delta t \left(D_2 \left(\mathbf{V}_x x^n + \mathbf{V}_{yy}^n \right) + g(\mathbf{U}^n, \mathbf{V}^n) \right) \quad (25)$$

where $*$ represents the element-wise multiplication of matrices. Our system is then discretized and numerically coded using MATLAB R2022a. The domain of simulation is on a rectangle $[-4, 4] \times [-4, 4]$ with the following initial conditions:

$$\begin{aligned} B(x, y, 0) &= 100e^{-0.75(x^2+y^2)}, \\ M(x, y, 0) &= \begin{cases} 0 & -1 < x, y < 1 \\ 2 & \text{otherwise} \end{cases} \\ M_i(x, y, 0) &= 0, \\ T(x, y, 0) &= 10(1 - e^{-0.01(x^2+y^2)}). \end{aligned} \quad (26)$$

The parameter values are taken from Table 1.

In our next numerical simulation, we show that an enhanced macrophage trafficking to the granuloma site has the potential to clearance of granuloma. We experimented with two infection sites with the following parameters and initial data. $\sigma_b = 0.17$; $\gamma_b = 6.5 \times 10^{-5}$; $\mu = 0.011$; $s = 0.005$; $\beta = 100$; $\gamma_i = 0.00017$; $\gamma_m = 0.011$; $\alpha = 1.1$; $\mu = 0.011$; $r = 4.5$; $a = 1500$; $\delta = 0.03$; $D_b = 0.1$, $D_m = 0.049$, $D_i = 0.049$, $D_T = 0.042$, $\chi = 0.045$, $\chi_T = 0.056$.

$$\begin{aligned} B(x, y, 0) &= \begin{cases} 10 & (x + 0.5)^2 + (y + 0.5)^2 < 1, \\ 10 & (x - 1)^2 + (y - 1)^2 < 1 \\ 0 & \text{otherwise} \end{cases} \\ M(x, y, 0) &= 10, \\ M_i(x, y, 0) &= 0, \\ T(x, y, 0) &= 10. \end{aligned} \quad (27)$$

The density of macrophage, T cells, bacteria, and infected macrophage is captured 1.25 days, 5 days, 10 days, and 12.5 days after initial infection. Numerical simulations reveal that the number of bacteria and infected macrophages continues to grow 12.5 days post-infection. However, with a stronger immune response when macrophages effectively migrate to the infection site, the numerical simulations show that the infection will be contained (Figures 8–10).

6 Discussion

The hallmark of tuberculosis (TB) is the formation of granuloma in the lung. A granuloma is composed of various immune cells with the infected macrophages at the center, and cytotoxic T cells at the outer layer. The immune response to the majority of Mtb infections is relatively successful, with most patients remaining asymptomatic, known as latent TB infection. The majority of granulomas will eventually sterilize, indicating that granulomas have the ability to clear infection.

In this paper, we developed a mathematical model for the formulation of granuloma in the lung. The model is described by a system of PDEs based on the classic chemotaxis model. We kept the dimension of the model low by only considering the most important cells during the process of the immune response against Mtb.

The qualitative analysis of the temporal model revealed that there is always an infection-free equilibrium that is locally asymptotically stable if the basic reproduction number is smaller than 1. Depending on certain conditions, there may exist a unique immune-free equilibrium as well in which case the bacteria persist and T cell count is zero. Finally, there may be zero, one, or two positive equilibria. When there are two positive equilibria, one is associated with a low number of Mtb, and the other is associated with a high number of Mtb.

The numerical simulations of the temporal model further revealed the outcome of the disease progression is closely related to a few parameters. In particular, we found that a higher engulfment rate by macrophages may lead to the clearance of the infection. While this result may not be surprising, what the other numerical simulation revealed is almost counter-intuitive. Under a certain parameter range, we found that a slightly higher infection rate γ_i can lead to the clearance of the virus.

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